Comptia N+

N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

DAY4

TASK:1

Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

RESULT

- Original Network: 192.168.10.0/24
- Required Subnets: 4
- Borrowed Bits = 2
- Subnet Mask = 255.255.255.192
- Hosts per Subnet = 62 usable hosts

Subnet	Network Address	Subnet Mask	Broadcast Address
Subnet 1	192.168.10.0	255.255.255.192 (/26)	192.168.10.63
Subnet 2	192.168.10.64	255.255.255.192 (/26)	192.168.10.127
Subnet 3	192.168.10.128	255.255.255.192 (/26)	192.168.10.191
Subnet 4	192.168.10.192	255.255.255.192 (/26)	192.168.10.255

TASKS:2

For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Make sure not to use the network or broadcast address for any device.

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RESULT

Selected Subnet

Subnet 1: 192.168.10.0/26

Usable IP Range: 192.168.10.1 – 192.168.10.62

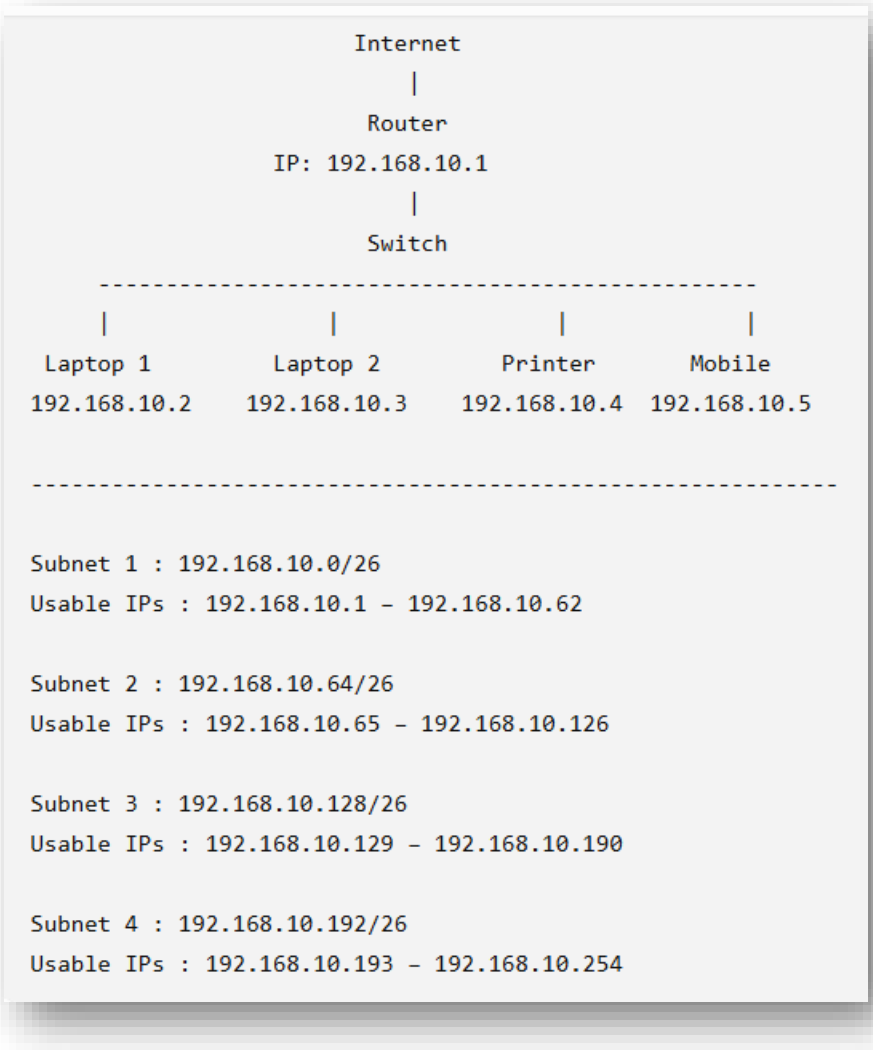
Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile Phone	192.168.10.5

TASKS:3

Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

RESULT

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TASKS:4

Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

RESULT

Step 1: Started with the network 192.168.10.0/24.

Step 2: I needed 4 subnets, so I borrowed 2 bits from the host portion because $2^2 = 4$ subnets.

Step 3: The subnet mask changed from /24 to /26, which is 255.255.255.192.

Step 4: Each subnet contains 64 IP addresses, but 2 addresses (network and broadcast) cannot be used. Therefore, each subnet has 62 usable host addresses.

Step 5: I listed the four subnet network addresses and broadcast addresses by increasing the network address by 64 each time:

- 192.168.10.0
- 192.168.10.64
- 192.168.10.128
- 192.168.10.192

Step 6: I selected the first subnet and assigned valid IP addresses to the router, laptops, printer, and mobile phone. I made sure not to use the network address or broadcast address.